

# SAT UZBEKISTAN

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A machine makes large boxes or small boxes, one at a time, for a total of 700 minutes each day. It takes the machine 10 minutes to make a large box or 5 minutes to make a small box. Which equation represents the possible number of large boxes,  $x$ , and small boxes,  $y$ , the machine can make each day?

- A.  $5x + 10y = 700$
- B.  $10x + 5y = 700$
- C.  $(x + y)(10 + 5) = 700$
- D.  $(10 + x)(5 + y) = 700$

A total of **364** paper straws of equal length were used to construct two types of polygons: triangles and rectangles. The triangles and rectangles were constructed so that no two polygons had a common side. The equation  $3x + 4y = 364$  represents this situation, where  $x$  is the number of triangles constructed and  $y$  is the number of rectangles constructed. What is the best interpretation of  $(x, y) = (24, 73)$  in this context?

- A. If **24** triangles were constructed, then **73** rectangles were constructed.
- B. If **24** triangles were constructed, then **73** paper straws were used.
- C. If **73** triangles were constructed, then **24** rectangles were constructed.
- D. If **73** triangles were constructed, then **24** paper straws were used.

An employee at a restaurant prepares sandwiches and salads. It takes the employee 1.5 minutes to prepare a sandwich and 1.9 minutes to prepare a salad. The employee spends a total of 46.1 minutes preparing  $x$  sandwiches and  $y$  salads. Which equation represents this situation?

- A.  $1.9x + 1.5y = 46.1$
- B.  $1.5x + 1.9y = 46.1$
- C.  $x + y = 46.1$
- D.  $30.7x + 24.3y = 46.1$

Keenan made **32** cups of vegetable broth. Keenan then filled  $x$  small jars and  $y$  large jars with all the vegetable broth he made. The equation  $3x + 5y = 32$  represents this situation. Which is the best interpretation of  $5y$  in this context?

- A. The number of large jars Keenan filled
- B. The number of small jars Keenan filled
- C. The total number of cups of vegetable broth in the large jars
- D. The total number of cups of vegetable broth in the small jars

$$4x + 3y = 24$$

Mario purchased 4 binders that cost  $x$  dollars each and 3 notebooks that cost  $y$  dollars each. If the given equation represents this situation, which of the following is the best interpretation of 24 in this context?

- A. The total cost, in dollars, for all binders purchased
- B. The total cost, in dollars, for all notebooks purchased
- C. The total cost, in dollars, for all binders and notebooks purchased
- D. The difference in the total cost, in dollars, between the number of binders and notebooks purchased

$$F = 2.50x + 7.00y$$

In the equation above,  $F$  represents the total amount of money, in dollars, a food truck charges for  $x$  drinks and  $y$  salads. The price, in dollars, of each drink is the same, and the price, in dollars, of each salad is the same. Which of the following is the best interpretation for the number 7.00 in this context?

- A. The price, in dollars, of one drink
- B. The price, in dollars, of one salad
- C. The number of drinks bought during the day
- D. The number of salads bought during the day

For a camping trip a group bought  $x$  one-liter bottles of water and  $y$  three-liter bottles of water, for a total of **240** liters of water. Which equation represents this situation?

- A.  $x + 3y = 240$
- B.  $x + y = 240$
- C.  $3x + 3y = 240$
- D.  $3x + y = 240$

A teacher is creating an assignment worth **70** points. The assignment will consist of questions worth **1** point and questions worth **3** points. Which equation represents this situation, where  **$x$**  represents the number of **1**-point questions and  **$y$**  represents the number of **3**-point questions?

A.  $4xy = 70$

B.  $4(x + y) = 70$

C.  $3x + y = 70$

D.  $x + 3y = 70$



Last week, an interior designer earned a total of \$1,258 from consulting for  $x$  hours and drawing up plans for  $y$  hours. The equation  $68x + 85y = 1,258$  represents this situation. Which of the following is the best interpretation of 68 in this context?

- A. The interior designer earned \$68 per hour consulting last week.
- B. The interior designer worked 68 hours drawing up plans last week.
- C. The interior designer earned \$68 per hour drawing up plans last week.
- D. The interior designer worked 68 hours consulting last week.

A gardener buys two kinds of fertilizer. Fertilizer A contains 60% filler materials by weight and Fertilizer B contains 40% filler materials by weight. Together, the fertilizers bought by the gardener contain a total of 240 pounds of filler materials. Which equation models this relationship, where  $x$  is the number of pounds of Fertilizer A and  $y$  is the number of pounds of Fertilizer B?

- A.  $0.4x + 0.6y = 240$
- B.  $0.6x + 0.4y = 240$
- C.  $40x + 60y = 240$
- D.  $60x + 40y = 240$

The equation  $40x + 20y = 160$  represents the number of sweaters,  $x$ , and number of shirts,  $y$ , that Yesenia purchased for \$160. If Yesenia purchased 2 sweaters, how many shirts did she purchase?

- A. 3
- B. 4
- C. 8
- D. 40

Tony spends \$80 per month on public transportation. A 10-ride pass costs \$12.50, and a single-ride pass costs \$1.50. If  $g$  represents the number of 10-ride passes Tony buys in a month and  $t$  represents the number of single-ride passes Tony buys in a month, which of the following equations best represents the relationship between  $g$  and  $t$ ?

- A.  $g + t = 80$
- B.  $g + t = 1.50 + 12.50$
- C.  $1.50g + 12.50t = 80$
- D.  $12.50g + 1.50t = 80$

$$5G + 45R = 380$$

At a school fair, students can win colored tokens that are worth a different number of points depending on the color. One student won  $G$  green tokens and  $R$  red tokens worth a total of 380 points. The given equation represents this situation. How many more points is a red token worth than a green token?

Figure A and figure B are both regular polygons. The sum of the perimeter of figure A and the perimeter of figure B is **63** inches. The equation  $3x + 6y = 63$  represents this situation, where  $x$  is the number of sides of figure A and  $y$  is the number of sides of figure B. Which statement is the best interpretation of **6** in this context?

- A. Each side of figure B has a length of **6** inches.
- B. The number of sides of figure B is **6**.
- C. Each side of figure A has a length of **6** inches.
- D. The number of sides of figure A is **6**.

$$2.5b + 5r = 80$$

The given equation describes the relationship between the number of birds,  $b$ , and the number of reptiles,  $r$ , that can be cared for at a pet care business on a given day. If the business cares for 16 reptiles on a given day, how many birds can it care for on this day?

- A. 0
- B. 5
- C. 40
- D. 80

A neighborhood consists of a **2**-hectare park and a **35**-hectare residential area. The **total** number of trees in the neighborhood is **3,934**. The equation  $2x + 35y = 3,934$  represents this situation. Which of the following is the best interpretation of  $x$  in this context?

- A. The average number of trees per hectare in the park
- B. The average number of trees per hectare in the residential area
- C. The **total** number of trees in the park
- D. The **total** number of trees in the residential area



Jay walks at a speed of **3** miles per hour and runs at a speed of **5** miles per hour. He walks for  $w$  hours and runs for  $r$  hours for a combined total of **14** miles. Which equation represents this situation?

- A.  $3w + 5r = 14$
- B.  $\frac{1}{3}w + \frac{1}{5}r = 14$
- C.  $\frac{1}{3}w + \frac{1}{5}r = 112$
- D.  $3w + 5r = 112$

A science teacher is preparing the 5 stations of a science laboratory. Each station will have either Experiment A materials or Experiment B materials, but not both. Experiment A requires 6 teaspoons of salt, and Experiment B requires 4 teaspoons of salt. If  $x$  is the number of stations that will be set up for Experiment A and the remaining stations will be set up for Experiment B, which of the following expressions represents the total number of teaspoons of salt required?

- A.  $5x$
- B.  $10x$
- C.  $2x + 20$
- D.  $10x + 20$

Each side of a **30**-sided polygon has one of three lengths. The number of sides with length **8 centimeters (cm)** is **5** times the number of sides  $n$  with length **3 cm**. There are **6** sides with length **4 cm**. Which equation must be true for the value of  $n$ ?

- A.  $5n + 6 = 30$
- B.  $6n + 6 = 30$
- C.  $8n + 3n + 4n = 30$
- D.  $8(5n) + 3n + 4(6) = 30$

Characteristics for Rock Types

| Rock type | Weight per volume<br>(lb/ft <sup>3</sup> ) | Cost per<br>pound |
|-----------|--------------------------------------------|-------------------|
| Basalt    | 180                                        | \$0.18            |
| Granite   | 165                                        | \$0.09            |
| Limestone | 120                                        | \$0.03            |
| Sandstone | 135                                        | \$0.22            |

A city is planning to build a rock retaining wall, a monument, and a garden in a park. The table above shows four rock types that will be considered for use in the project. Also shown for each rock type is its weight per volume, in pounds per cubic foot (lb/ft<sup>3</sup>), and the cost per pound, in dollars. Only basalt, granite, and limestone will be used in the garden. The rocks in the garden will have a total weight of 1,000 pounds. If 330 pounds of granite is used, which of the following equations could show the relationship between the amounts,  $x$  and  $y$ , in ft<sup>3</sup>, for each of the other rock types used?

- A.  $165x + 180y = 670$
- B.  $165x + 120y = 1,000$
- C.  $120x + 180y = 670$
- D.  $120x + 180y = 1,000$

$$x + y = 75$$

The equation above relates the number of minutes,  $x$ , Maria spends running each day and the number of minutes,  $y$ , she spends biking each day. In the equation, what does the number 75 represent?

- A. The number of minutes spent running each day
- B. The number of minutes spent biking each day
- C. The total number of minutes spent running and biking each day
- D. The number of minutes spent biking for each minute spent running

A salesperson earns 2 points for each call made to a potential customer and 8 points for each visit made to a potential customer. Last week, the salesperson earned a total of 732 points. The equation  $2x + 8y = 732$  represents this situation. Which is the best interpretation of  $8y$  in this context?

- (A) The number of calls the salesperson made to potential customers last week
- (B) The number of visits the salesperson made to potential customers last week
- (C) The total number of points the salesperson earned from calls made to potential customers last week
- (D) The total number of points the salesperson earned from visits made to potential customers last week



A certain store earned total of \$1700 this year, and the equation  $10x + 15y = 1700$  represents the situation, where  $x$  is the number of short-sleeved shirts and  $y$  is the number of long-sleeved shirts the store sold during the year. What is the best interpretation of 15 in the equation?

- A) Number of short-sleeved shirts
- B) Number of long-sleeved shirts
- C) Price of each short-sleeved shirt

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- D) Price of each long-sleeved shirt

The equation  $0.95c + 0.05n = 8.87$  represents the density of a copper-zinc alloy, where  $c$  is the density, in grams per cubic centimeter ( $\text{g}/\text{cm}^3$ ), of copper,  $n$  is the density, in  $\text{g}/\text{cm}^3$ , of zinc, and  $8.87 \text{ g}/\text{cm}^3$  is the density of the alloy. The density of copper is  $8.96 \text{ g}/\text{cm}^3$ . What is the density of zinc, in  $\text{g}/\text{cm}^3$ ?

- A) 0.09
- B) 0.47
- C) 7.16
- D) 8.51



$$19.5x + 24.25y = 583$$

Odalys ordered mulch and river rock, which cost a total of \$583, for her home. The given equation represents the relationship between the number of cubic yards of mulch,  $x$ , and the number of tons of river rock,  $y$ , Odalys ordered. How much more, in dollars, did a ton of river rock cost Odalys than a cubic yard of mulch?

$$C = 24f + 88e$$

The cost in collars,  $C$ , from an airplane flight given that the average cost of a first class seat is  $f$  dollars and the average cost of an economy seat is  $e$  dollars is given by the equation if all seats are sold. What is the best interpretation of  $88e$  as shown in the equation?

Choose 1 answer:

- ☐ Ⓐ There are  $88e$  economy seats on the flight.
- ☐ Ⓑ The total cost due to economy seats is  $88e$  dollars.
- ☐ Ⓒ The increase in total cost for each economy seat sold is  $88e$  dollars.
- ☐ Ⓓ The increase in total cost is  $88e$  dollars when economy seat prices increase by 1 dollar.

A neighborhood consists of a 2-hectare park and a 35-hectare residential area. The total number of trees in the neighborhood is 3,934. The equation  $2x + 35y = 3,934$  represents this situation. Which of the following is the best interpretation of  $x$  in this context?

- A) The average number of trees per hectare in the park
- B) The average number of trees per hectare in the residential area
- C) The total number of trees in the park
- D) The total number of trees in the residential area

A geographer found that the land area of Aruba is 75 square miles and the land area of Bermuda is 19 square miles. Based on these data, the geographer used the equation  $75x + 19y = T$  to estimate the total number of residents,  $T$ , of the two islands. Which is the best interpretation of  $x$  in this context?

- A) The total number of residents of Aruba
- B) The number of square miles of Aruba
- C) The total number of residents of both islands
- D) The population density, in residents per square mile, of Aruba.



A school club is paid \$315 for recycling  $c$  pounds of aluminum cans and  $p$  tons of paper. The equation  $0.50c + 60p = 315$  represents this situation. What is the interpretation of  $60p$  in this context?

- A) The total weight, in tons, of paper the club recycled
- B) The total amount, in dollars, the club is paid for each ton of paper recycled
- C) The total weight, in tons, of all the aluminum cans and paper the club recycled
- D) The total amount, in dollars, the club is paid for  $p$  tons of paper recycled

A certain apprentice has enrolled in **85** hours of training courses. The equation  **$10x + 15y = 85$**  represents this situation, where  **$x$**  is the number of on-site training courses and  **$y$**  is the number of online training courses this apprentice has enrolled in. How many more hours does each online training course take than each on-site training course?

A store sells two different-sized containers of a certain Greek yogurt. The store's sales of this Greek yogurt totaled **1,277.94** dollars last month. The equation  **$5.48x + 7.30y = 1,277.94$**  represents this situation, where  **$x$**  is the number of smaller containers sold and  **$y$**  is the number of larger containers sold. According to the equation, which of the following represents the price, in dollars, of each smaller container?

- A. **5.48**
- B.  **$7.30y$**
- C. **7.30**
- D.  **$5.48x$**

$$3a + 4b = 25$$

A shipping company charged a customer \$25 to ship some small boxes and some large boxes. The equation above represents the relationship between  $a$ , the number of small boxes, and  $b$ , the number of large boxes, the customer had shipped. If the customer had 3 small boxes shipped, how many large boxes were shipped?

- A. 3
- B. 4
- C. 5
- D. 6



A school classroom with a total of 4,200 floor tiles is divided into a 30 square foot lab area and an 80 square foot seating area. The number of tiles on the entire classroom floor can be represented by the equation  $30a + 80b = 4,200$ . In this context, which of the following does  $b$  represent?

- ☐ Ⓐ The average number of tiles per square foot in the lab area
- ☐ Ⓑ The total number of tiles in the lab area
- ☐ Ⓒ The average number of tiles per square foot in the seating area
- ☐ Ⓓ The total number of tiles in the seating area

Several tiles labeled with either an A or a B are placed in a bag, and tiles are worth a different point value depending on the label. The equation  $15a + 10b = 100$  represents the situation when  $a$  of the A tiles and  $b$  of the B tiles are drawn from the bag for a total of 100 points. How many points would be earned by drawing one A tile and one B tile from the bag?

**Answer Preview:**

A carpenter hammers 10 nails per minute and installs 7 screws per minute during a project. Which of the following equations represents the scenario if the carpenter hammers nails for  $x$  minutes, installs screws for  $y$  minutes, and uses a combined total of 200 nails and screws?

Ⓐ  $\frac{1}{10}x + \frac{1}{7}y = 200$

Ⓑ  $\frac{1}{10}x + \frac{1}{7}y = 3,420$

Ⓒ  $10x + 7y = 200$

Ⓓ  $10x + 7y = 3,420$

A fruit stand sells a total of 200 apples and bananas. The apples are sold in bags of 4 apples per bag, and the bananas are sold in bunches of 6 bananas each. Which of the following equations best represents the number of bags of apples,  $a$ , and bunches of bananas,  $b$ , that could be sold at the fruit stand?

☐ (A)  $(a + b)(4 + 6) = 200$

☐ (B)  $(4 + a)(6 + b) = 200$

☐ (C)  $4a + 6b = 200$

☐ (D)  $6a + 4b = 200$

Last week, a softball equipment company made \$26,000 from the sale of equally-priced bats and equally-priced gloves. When  $b$  is the number of bats sold and  $g$  is the number of gloves sold, the equation  $500b + 300g = 26,000$  represents this situation. The sales price of each glove is how many dollars less than the sales price of each bat?

**Answer Preview:**

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Mark for Review

A record collector is looking to buy  $c$  classic albums and  $m$  modern albums. The equation  $35c + 20m = 750$  represents this situation if the record collector spends a total of \$750 on albums. How much more, in dollars, does one classic album cost than one modern album?

**Answer Preview:**

A model builder is constructing  $b$  of the same big ship model and  $s$  of the same small ship model. Each big ship model has 150 parts, and each small ship model has 80 parts. If the model builder uses a total of 2,600 parts, which of the following equations could represent the number of big and small ship models the model builder constructs?

(A)  $(b + s)(150 + 80) = 2,600$

(B)  $(150 + b)(80 + s) = 2,600$

(C)  $150b + 80s = 2,600$

(D)  $80b + 150s = 2,600$



The equation  $0.75s + 2.25l = 33$  models the relationship between the number of short tours,  $s$ , and the number of long tours,  $l$ , that the members of a museum staff lead on a single day. If the members of the museum staff lead 5 short tours on a single day, what is the maximum number of long tours they can lead on that day?

(A) 2.25

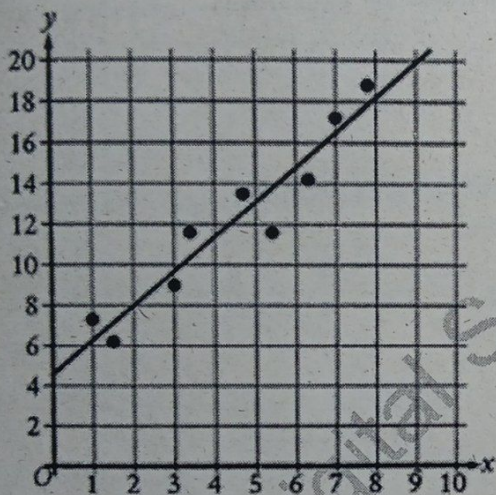
(B) 13

(C) 29

(D) 33

Antonia spent a total of \$108 at the art supply store. She purchased  $s$  small tubes of oil paint and returned  $l$  large tubes of oil paint for a refund. Her purchase can be represented by the equation  $12s - 24l = 108$ . Which of the following is the best interpretation of  $24l$  in this context?

- (A) The number of small tubes Antonia purchased
- (B) The number of large tubes Antonia returned
- (C) The total amount of money Antonia spent on purchasing the small tubes
- (D) The total amount of money Antonia received for returning the large tubes



The scatterplot shows the relationship between two variables,  $x$  and  $y$ . A line of best fit for the data is also shown. For how many of the 9 data points is the actual  $y$ -value less than the  $y$ -value predicted by the line of best fit?

- A) 2
- B) 4
- C) 5
- D) 7